

TIMOTIUS WILLIAM WUISANG

Engineering Physics Fresh Graduate | Research Assistant | R&D, Data & Analytics, IoT Automation & Building Physics Enthusiast
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PROFESSIONAL SUMMARY

Engineering Physics fresh graduate from Universitas Gadjah Mada with experience in research support, Python-based data analysis, basic informatics teaching, and IoT-oriented engineering concept design. Interested in R&D, building physics and energy efficiency, AI/data science, instrumentation, and applied engineering projects.

EDUCATION

Bachelor of Engineering in Engineering Physics | Universitas Gadjah Mada | Yogyakarta 2022 - 2026

- GPA: 3.57/4.00
- Relevant interests: building physics, energy efficiency, AI/data science, IoT, instrumentation, and material characterization.

EXPERIENCE

Research Assistant - Thermochromic Transparent Wood Project | UGM | Yogyakarta Feb - Jul 2026

- Supported lecturer-led research on transparent wood with leuco dye-based thermochromic coating for optical and thermal performance studies.
- Assisted experimental work involving delignification, polymer impregnation, polyurethane coating preparation, and spray coating variation.
- Analyzed and summarized experimental data, including CIELAB color change, luminance transmittance, response time, and thermal conductivity.
- Prepared literature review notes and technical documentation on transparent wood, thermochromic materials, and passive building applications.

Research Intern - Research Division | YKK AP R&D Center | Tangerang Dec 2024 - Jan 2025

- Conducted a preliminary study and design concept for an IoT-based automated window ventilation system supporting passive building strategies.
- Contributed to technical discussions and co-authored a collaborative research proposal related to building energy efficiency.

Basic Informatics Laboratory Assistant | UGM | Yogyakarta Aug-Dec 2024

- Assisted students during basic informatics practicum sessions using Python.
- Evaluated student assignments and practicum reports while providing feedback on programming logic and report clarity.

AI Project Participant | Intel Prakarsa Muda 2021 - Sagasitas Indonesia | Yogyakarta Jun-Nov 2021

- Collaborated in a team to develop an AI-based technology project and communicate the project concept to mentors.

PROJECTS

E-commerce Shipping Prediction - Built a machine-learning workflow for shipment prediction using Python, Pandas, visualization, and Scikit-learn; achieved 69% accuracy.
kaggle.com/code/timotiuswilliamw/e-commerce-shipping-prediction-69-accuracy

E-commerce Shipping Exploratory Data Analysis - Explored shipment-related variables and communicated initial findings through simple visual analysis.
kaggle.com/code/timotiuswilliamw/e-commerce-shipping-very-simple-eda-beginner

TRAINING

Data Science Training | KMTETI FT UGM Workshop Division | Yogyakarta Sep-Nov 2023

- Completed introductory training in data processing, visualization, and machine-learning workflow fundamentals.

SKILLS

Engineering & Research: Experimental testing, process monitoring, material characterization, heat transfer, technical documentation, laboratory troubleshooting

Embedded Systems & Hardware: Microcontroller Programming (Arduino, ESP32), Circuit Design & Troubleshooting, Sensor Integration (Temperature, Humidity, Air Velocity), Actuator Control

Data & Digital: Python, Pandas, Matplotlib, Seaborn, Scikit-learn, Microsoft Excel, data cleaning, exploratory analysis, data visualization

Professional: Analytical problem solving, student mentoring, teamwork, technical reporting, presentation, adaptability, willingness to learn

Languages: Indonesian (Native), English (Intermediate)